

The background of the slide is a dark, teal-toned image featuring various financial market data visualizations. In the upper left, there are several candlestick charts. Across the middle and lower sections, there are multiple line graphs with different colors (light blue, orange, and dark blue) showing fluctuating trends. The overall aesthetic is high-tech and data-oriented.

Predicting Economic Trends:

A Data-driven approach to GDP Trends

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Gross Domestic Product and the importance of its prediction accuracy



Government Policy & Interest Rates: Central Banks through GDP predictions can decide whether to decrease or increase interest rates.



Business Investment & Expansions: Companies and investors may decide to invest in or pull out of a specific region based on GDP predictions.

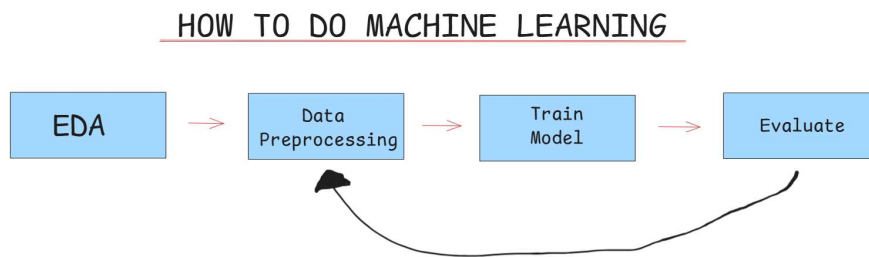


Budgeting & Public Services: Government based on the GDP predictions can increase or decrease funds to various sectors.

Project Overview

- Project goal to predict accurately the GDP evolution of countries and geographic regions
- 150+ countries examined
- World Bank data of 60+ years used (1960-2023)
- Four different Machine Learning (ML) models used
- Data split into training and testing periods with 80:20 ratio (training: 1960-2010; testing: 2010-2023)
- Forecast generated for 5 years in the future

A four-step data process leveraging Machine Learning



A four-step process

1. Exploratory Data Analysis (EDA)
2. Data Cleaning and Preprocessing
3. Model Selection & Training
4. Results & Evaluation

Dataset Composition

A total of ~ 14k GDP data points including 250+ countries and geographic regions

```
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RangeIndex: 13979 entries, 0 to 13978
Data columns (total 6 columns):
#   Column          Non-Null Count  Dtype  
---  -
0   Country Name    13979 non-null object  
1   Country Code    13979 non-null object  
2   Year            13979 non-null int64   
3   Value           13979 non-null float64  
4   Population       13979 non-null float64  
5   GDP per Capita  13979 non-null float64  
dtypes: float64(3), int64(1), object(2)
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GDP (Target Variable)

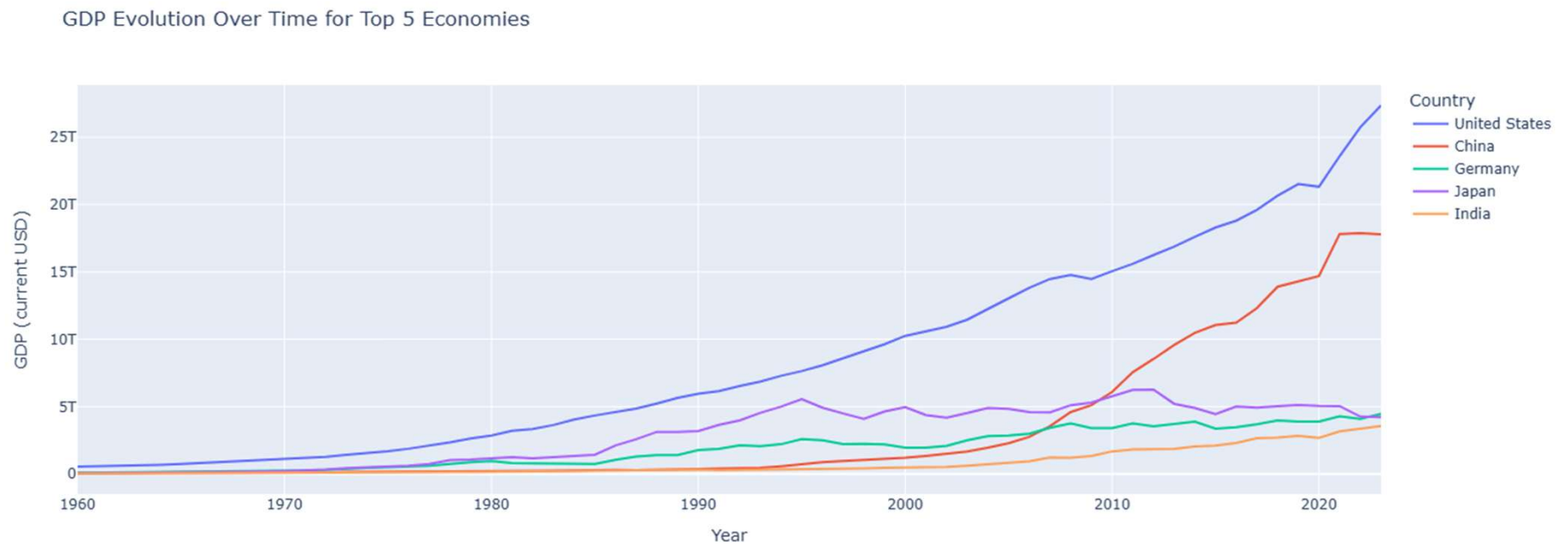
Additional Information

	Country Name	Country Code	Year	Value	Population	GDP per Capita
0	Afghanistan	AFG	2000	3.521418e+09	20130327.0	174.930991
1	Afghanistan	AFG	2001	2.813572e+09	20284307.0	138.706822
2	Afghanistan	AFG	2002	3.825701e+09	21378117.0	178.954088
3	Afghanistan	AFG	2003	4.520947e+09	22733049.0	198.871116
4	Afghanistan	AFG	2004	5.224897e+09	23560654.0	221.763654

	Country Name	Country Code	Year	Value	Population	GDP per Capita
13974	Zimbabwe	ZWE	2019	2.183223e+10	15271368.0	1429.618809
13975	Zimbabwe	ZWE	2020	2.150970e+10	15526888.0	1385.319351
13976	Zimbabwe	ZWE	2021	2.837124e+10	15797210.0	1795.965152
13977	Zimbabwe	ZWE	2022	2.736663e+10	16069056.0	1703.063774
13978	Zimbabwe	ZWE	2023	2.653827e+10	16340822.0	1624.047646

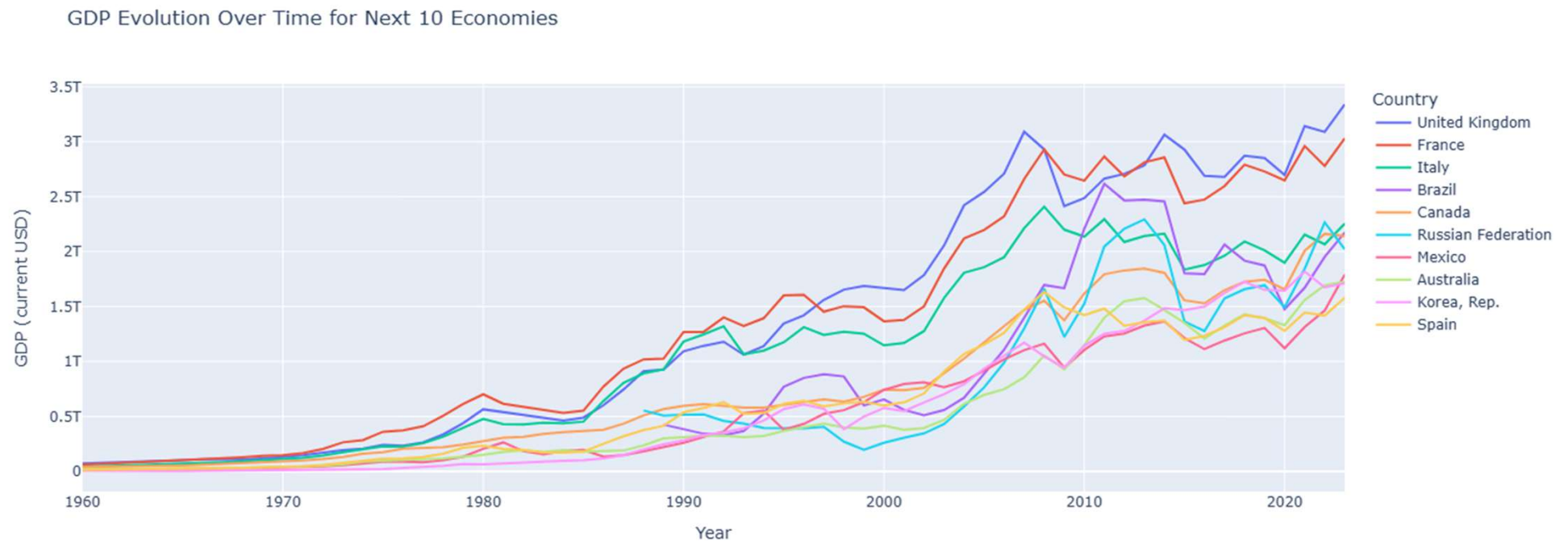
GDP Evolution (I): 5 largest world economies

China and the US experience much faster GDP growth compared to other key countries

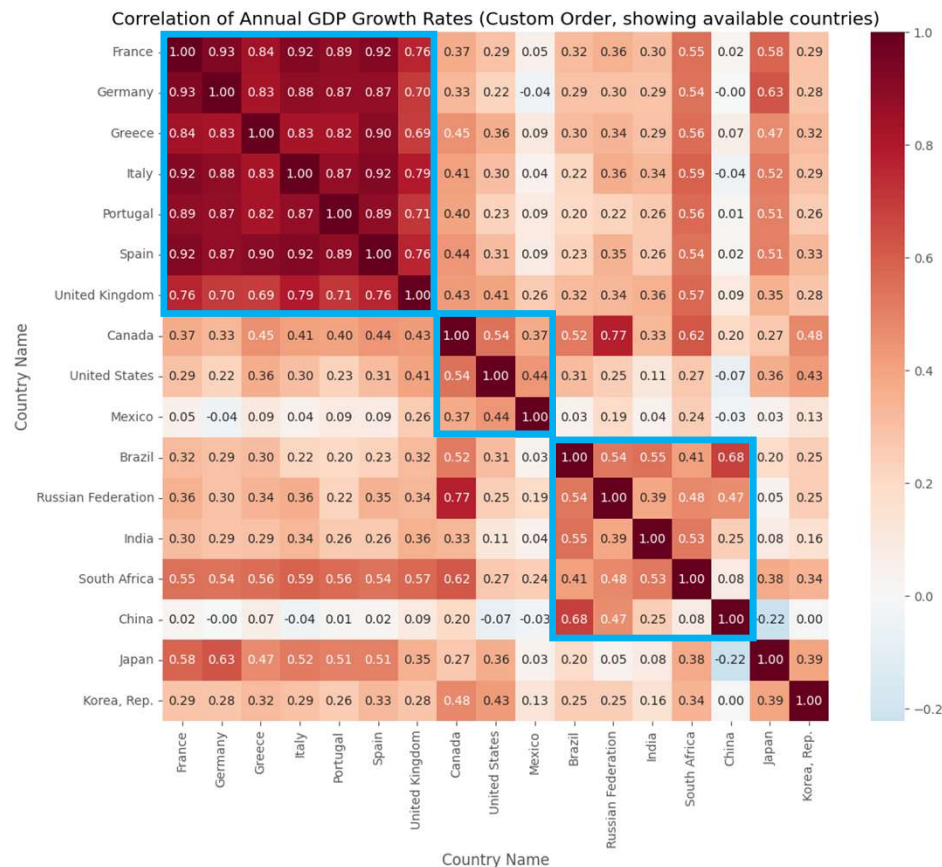


GDP evolution (II): Next 10 largest economies

UK and France with larger GDP than the rest although gap is closing



EDA: Building the correlation matrix for key countries



Key conclusions :

- Countries within major economic blocks have positive GDP correlations
- EU countries display significant positive correlation
- NAFTA countries display significant positive correlation
- China with negative correlation vs. most Western countries, but also with Japan and South Korea
- BRICS economies positively correlated

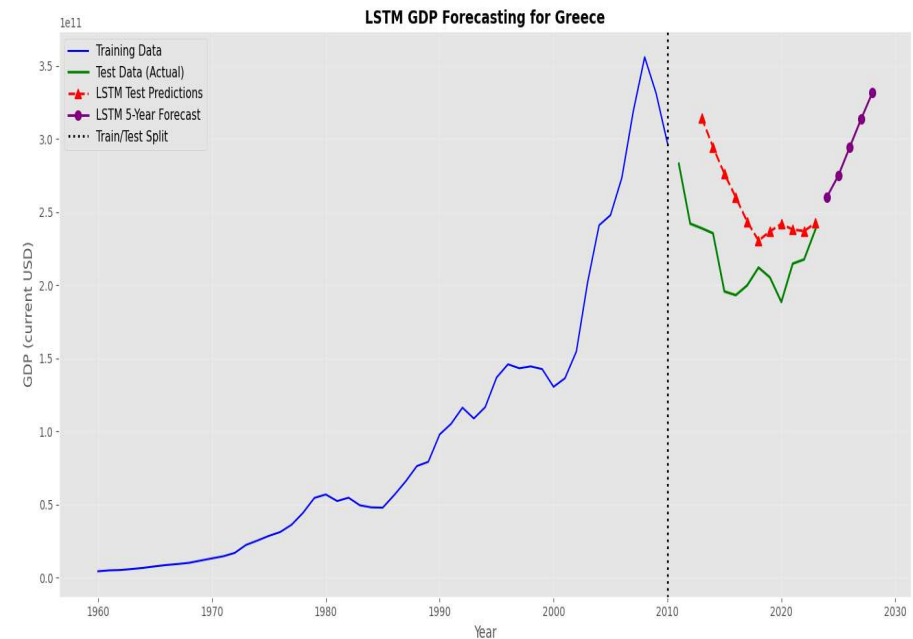
Model Selection and Training

4 ML models used to predict the future value of the GDP of countries

Linear Regression	ARIMA	Prophet	LSTM
• Models GDP as a straight-line trend over time • Simple and interpretable, but assumes a constant rate of growth	• Captures trends and short-term patterns by combining past GDP values and forecast errors • Well-suited for stationary time series	• Decomposes GDP into trend and seasonality components • Robust to missing data and handles irregular fluctuations well	• Recurrent neural network that learns complex, long-term dependencies in GDP data • Most powerful but requires more data and tuning • Runs slower

Greece: LSTM with best predicting performance albeit with MAPE 20%

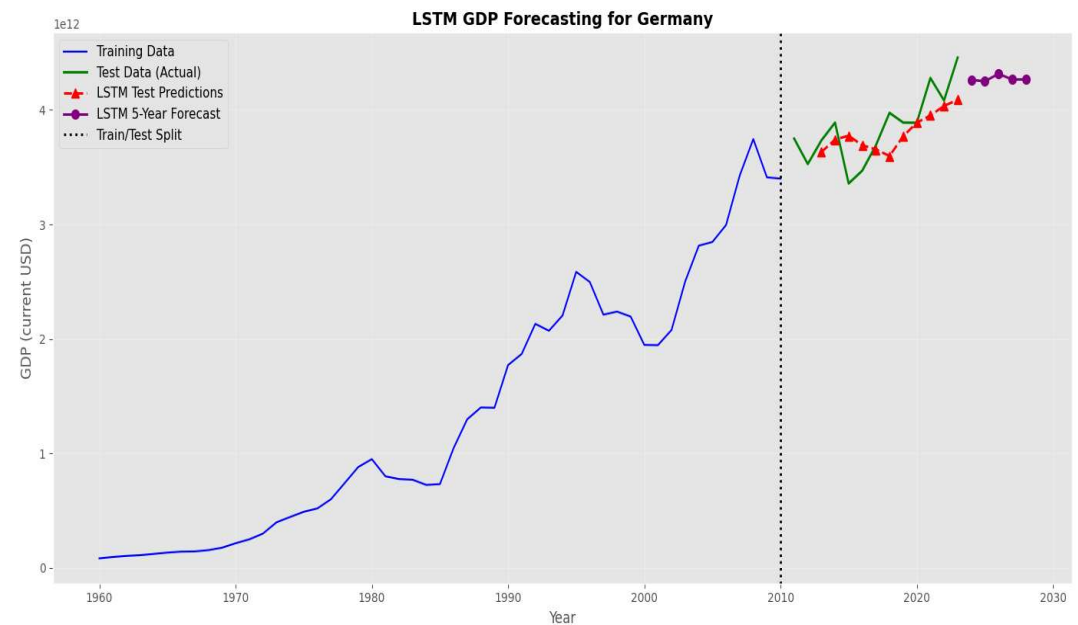
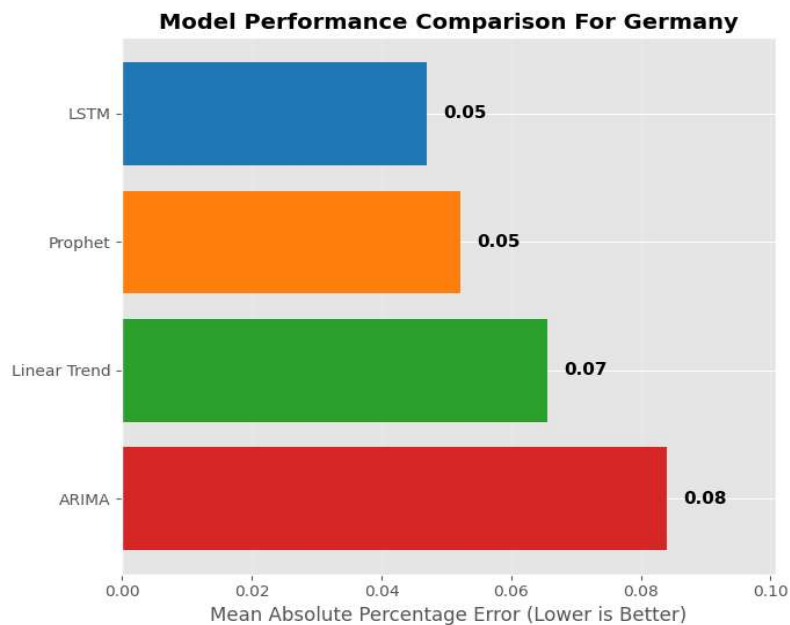
Impacted by the “black swan” event of economic collapse



LSTM accurately predicts the downward trend for Greece's GDP but not its magnitude and expects high growth for the five years ahead

Germany: LSTM with best predicting performance albeit narrowly

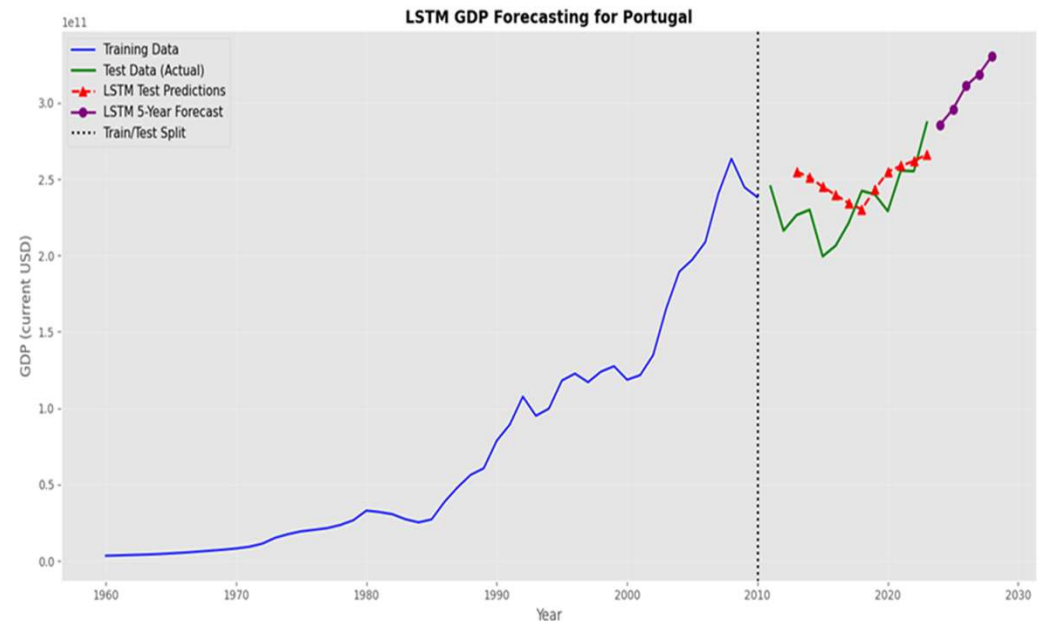
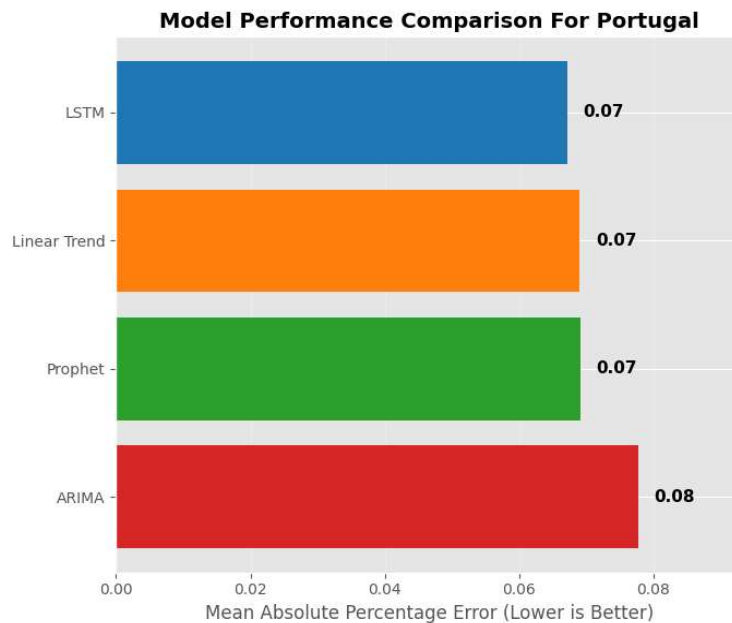
Germany's economy seems to be easily predictable



LSTM accurately predicts the upward trend for Germany's GDP and expects the economy to remain stationary for the five years ahead

Portugal: LSTM with best predicting performance with Linear Regression second

Identical error rates confirm Portugal's GDP highly predictable trajectory



LSTM accurately predicts the upward trend for Portugal's GDP and expects high growth for the five years ahead

Limitations and Improvements

Challenges Identified and Opportunities for enhancement

Limitations
Ignores the “Real world” • Wars • Elections Vulnerability to “Black Swan” events • COVID – 19 • 2008 economic crisis • Oil price spikes

Potential Improvements
Add more advanced models • Prophet with use of MCMC • Other neural networks Add variables to improve model accuracy • Unemployment • Inflation • Population growth

The background of the slide is a dark, teal-colored image featuring a complex financial chart. It includes a candlestick chart at the top, several overlapping line graphs in light blue and orange, and a bar chart at the bottom. Faint, out-of-focus numbers are visible in the background, suggesting a data-rich environment.

THANK YOU ! ☺
